

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Second Semester of M. Tech (Electrical Power System) Examination****May 2018****EE746 ADVANCED POWER SYSTEM PROTECTION****Date: 05.05.2018, Saturday****Time: 01:30 p.m. To 04:30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q – 1** Explain desirable attributes of protective relay. **[07]****Q – 2.a Attempt any one.** **[08]**

1. For CCVT attempt the following, with explanation of nomenclature required:
 1. Draw equivalent circuit diagram of CCVT as seen from primary side of VT.
 2. Draw schematic diagram of CCVT with RF connection used for power line communication.
 3. Mention the role of tuning and drainage reactor.
 4. Derive the transient response of CCVT and hence draw the typical transient occurring for fault voltage.
2. Mentioned the equation of ϕ_{ac} & ϕ_{dc} for the CT excited with AC and DC current and hence derive the equation for CT Oversizing Factor. Mention meaning of all nomenclature used in equation and derivation.

Q – 2.b Attempt all.

1. Distinguish between PSM and TMS of a relay. **[03]**
2. Compare of protection and measurement CT. Also draw the characteristics of both CT. **[03]**

OR**Q – 2.b Attempt all.**

1. How can CCVT be used in Pilot wire communication? **[03]**
2. What are the reasons that required for new measurement and monitoring technology in power system? **[03]**

Q - 3 Attempt any one. **[14]**

- a. A radial feeder is shown in Figure 1. The CT ratio and maximum and minimum fault currents at various substations A, B, C, D and E are indicated in the table and full-load currents of feeders are indicated in Figure. Determine the plug-settings and time-settings of standard IDMT relays used for this feeder protection. TMS of R4 is set at 0.1. Allow for 10% overload.

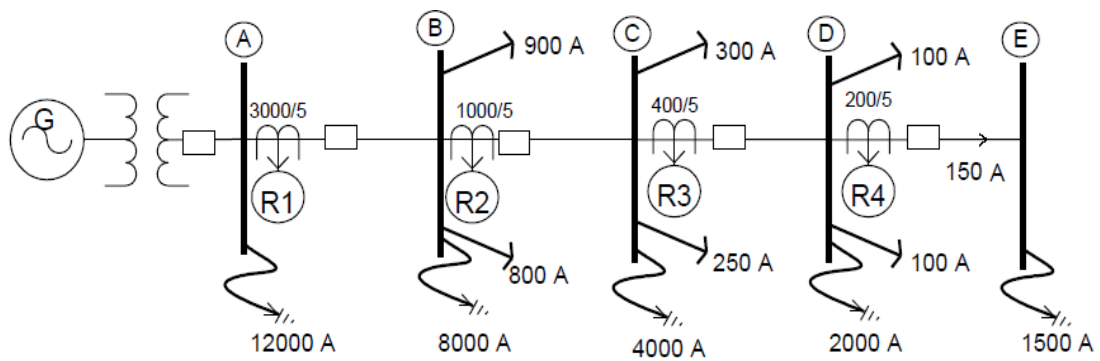


Fig. 1

- b. Find the primary backup relay coordination for relay R_1 and R_4 from figure 2. Using LINKNET structure.

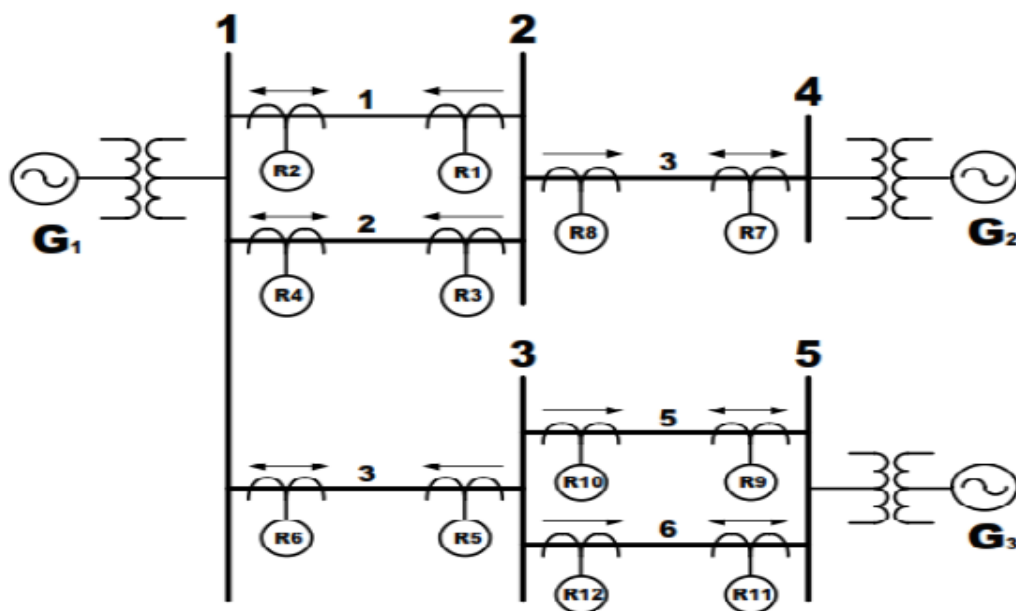


Fig. 2

SECTION – II

Q - 4 Attempt all.

- List out various component in block diagram of Numerical relay. [01]
- Explain any three difference between P class and M class Phasor Measurement Units. [03]
- Derive relation for estimation of Phasor using two sample technique. [04]
- Define and Explain the term Synchro Phasor as per IEEE standard. [02]

Q – 5.a Attempt any one.

[06]

- Explain the concept of moving window and hence explain how to calculate time varying Fundamental Phasor with the help of Fourier Transform.
- Mathematically, prove the following statement.
“ The Fourier Transform based Phasor estimation gives wrong result when non integer multiple of fundamental frequency component are present in signal”.

Q – 5.b Attempt any one.**[10]**

1. Derive the equation to estimate the phasor using Least square error estimation technique. Also prove the resultant equations are similar to Fourier Transform equation for fundamental Phasor estimation.
2. Derive the mathematical relation between Fourier Transform and Walsh Transform & Hence state equation to find out fundamental Phasor component.

Q – 6.a Attempt any one.**[05]**

1. Draw the block diagram for Phasor Measurement Unit and explain the working in brief.
2. Calculate $Wal(5,4)$ for 16 coefficients/samples of Walsh showing all necessary mathematical steps.

Q – 6.b Attempt any one.**[04]**

1. Draw all the waveform of Walsh coefficient for total 8 samples in a Walsh signal.
2. Discuss various online and Off line applications of PMU.
